

New Experiments - Findings Addendum

Tile-level redundancy in LLMs. Qwen3-4B (instruct) and a second model, Llama-3.2-3B-Instruct. Tile size 32x32. Downstream measured on HellaSwag / PIQA / ARC-Easy; retained = (acc - chance) / (acc_dense - chance).

Standalone addendum to FINDINGS_clean, which is left unchanged. Covers four experiments run to close the report's open next-steps: (A) the recommended-method downstream ladder, (B) iterative calibration, (C) non-square strips, and (D) the second-model generality gate. Draft for review; numbers are first-pass where noted.

A. Recommended-method downstream ladder: perplexity inverts real capability

What we test

Whether the near-dense perplexity ordering among the three repair variants (random_recon, wanda_recon, sparsegpt_recon) is mirrored by real downstream capability, across the capability-relevant band (5 / 10 / 20 percent structured sparsity).

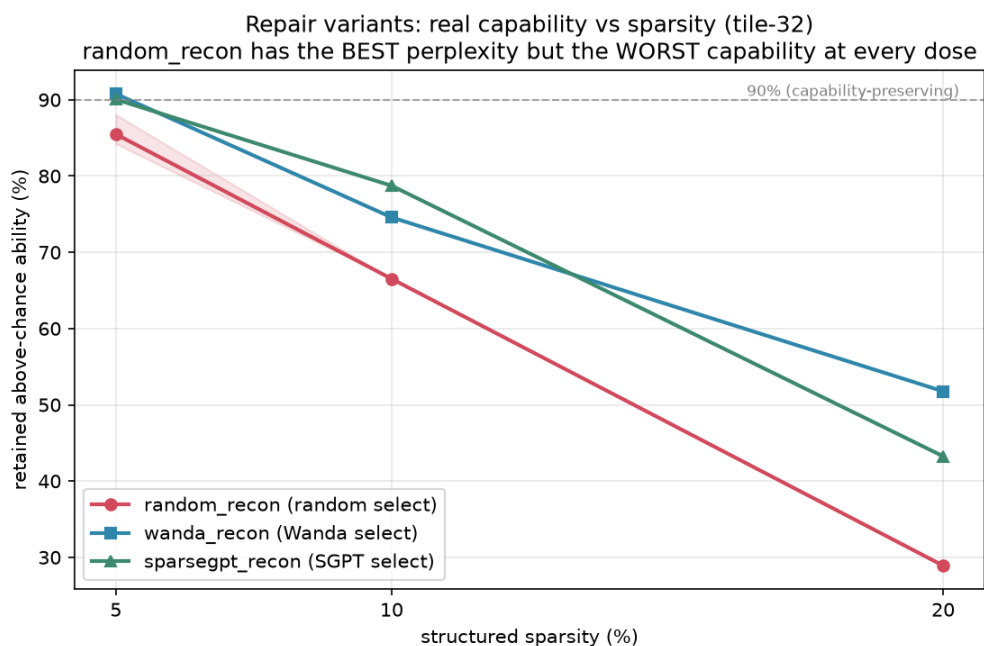
Why we ran it

The report noted this downstream data was unfinished. If methods that all look near-dense on perplexity disagree on real tasks, ranking repair methods by perplexity is unsafe.

Result

They disagree - perplexity inverts capability. random_recon has the best (lowest) whole-model perplexity at every dose, but the worst retained ability at every dose. And given repair, calibrated selection beats random selection by a margin that GROWS with sparsity.

variant (repair)	p5	p10	p20	whole-model ppl
random_recon	85	67	29	14.6 / 16.5 / 24.4
wanda_recon	91	75	52	15.4 / 18.4 / 24.0
sparsegpt_recon	90	79	43	15.6 / 18.8 / 30.9



Reliability

Downstream-anchored (full task sets). random_recon is 5 seeds at p5, single median seed at p10/p20; the calibrated variants are deterministic. This QUALIFIES Finding 4's 'selection contributes nothing' - that claim is perplexity-scoped; on real capability, selection matters, and the gap widens from ~6 pp at p5 to ~14-23 pp at p20.

B. Iterative (least-squares-only) calibration does not raise the ceiling

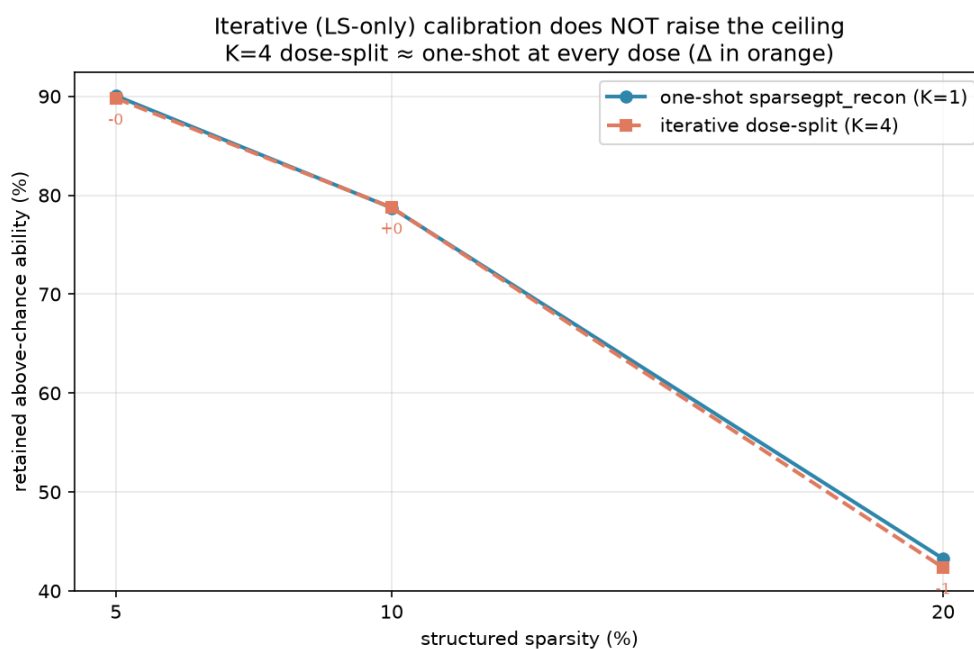
What we test

Whether splitting the target sparsity into K increments and re-collecting the calibration signal on the progressively-pruned model between increments (iterative / sequential calibration) lifts the ~5 percent structured ceiling above one-shot pruning.

Result

It does not. A K=1 run reproduces one-shot exactly (30.95 ppl at 20 percent, which validates the implementation). K=4 dose-splitting is statistically indistinguishable from one-shot:

dose	iterative K=4	one-shot	delta
p5	90%	90%	0
p10	79%	79%	0
p20	42%	43%	-1



Reliability

Downstream-anchored; the K=1 == one-shot anchor confirms correctness. Mechanism: the one-shot pipeline ALREADY re-collects per-layer calibration on the progressively-pruned model, and the repair is already exact for its mask, so least-squares-only iteration adds no new information. Raising the ceiling would require gradient information (light fine-tuning between cuts) - a larger-scope direction. This closes the LS-only iterative fork with a clean negative result.

C. Non-square strips do not rescue redundancy; the '1xN' orientation is backwards

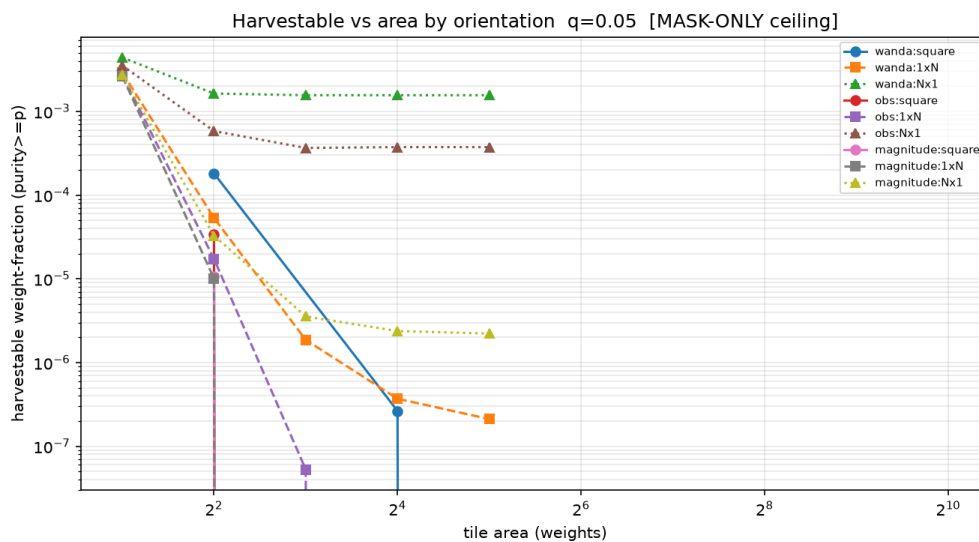
What we test

Whether non-square 1xN (wide) or Nx1 (tall) weight strips harvest more of the diffuse redundancy than square tiles of equal area, and which orientation the row anisotropy favors.

Result

Strips do not rescue usable redundancy. At the 5 percent redundant set (purity ≥ 0.9), the best strip harvests less than 0.5 percent of weights, versus ~ 5 percent at 1x1 and 0.00 percent at square 32x32 - the redundancy is diffuse even at the two-element scale. Of the two orientations, Nx1 (tall, spanning output neurons) marginally beats 1xN (wide) - the OPPOSITE of the '1xN row strips' the report proposed.

orientation	best shape	harvestable
Nx1 (tall)	2x1	0.44%
1xN (wide)	1x2	0.29%
square	32x32	0.00%



Reliability

Weight-space probe (mask-only ceiling), CPU-validated on planted structure. Caveat: this run's aggregate anisotropy came out near-isotropic ($A \sim 1.05$), not reproducing the original probe's gate_proj-driven $A \sim 0.24$; the harvestable measurement (the direct arbiter) is unaffected, but the anisotropy metric should be reconciled before it is cited. Practical verdict: do not pursue non-square strips.

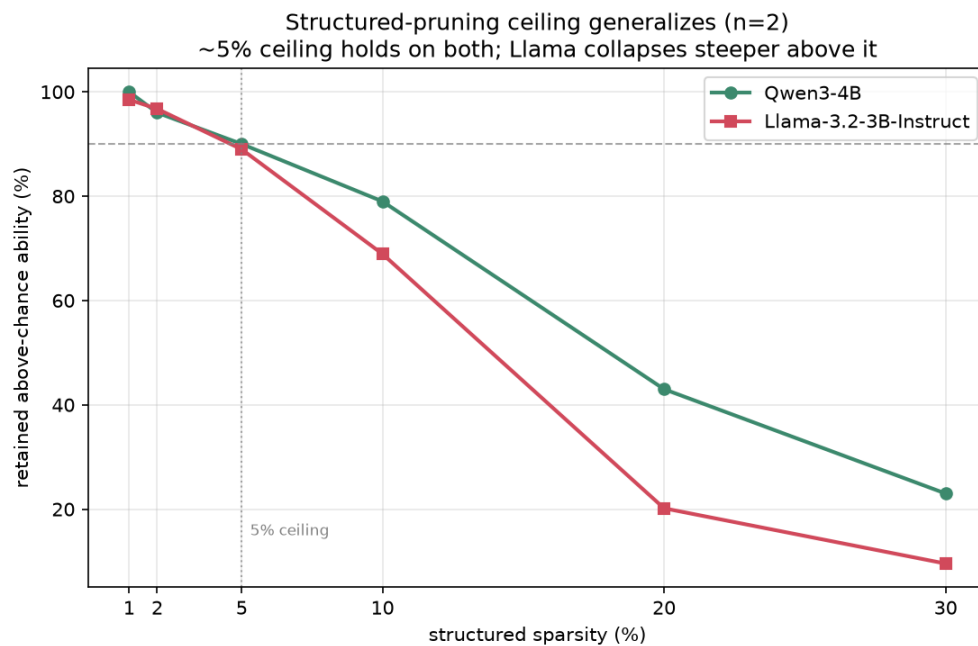
D. Second model (Llama-3.2-3B-Instruct): what transfers and what does not

What we test

Whether the capability-critical findings replicate on a second architecture. Model: Llama-3.2-3B-Instruct, matched to Qwen3-4B (also an instruct model), same tile-32 and same WikiText + HellaSwag/PIQA/ARC-Easy setup. Minimum set: F1 (structured ladder), F4 (repair vs selection), F10 (structured tax). Llama dense: acc 0.716/0.768/0.709, ppl 10.45 (Qwen 13.22).

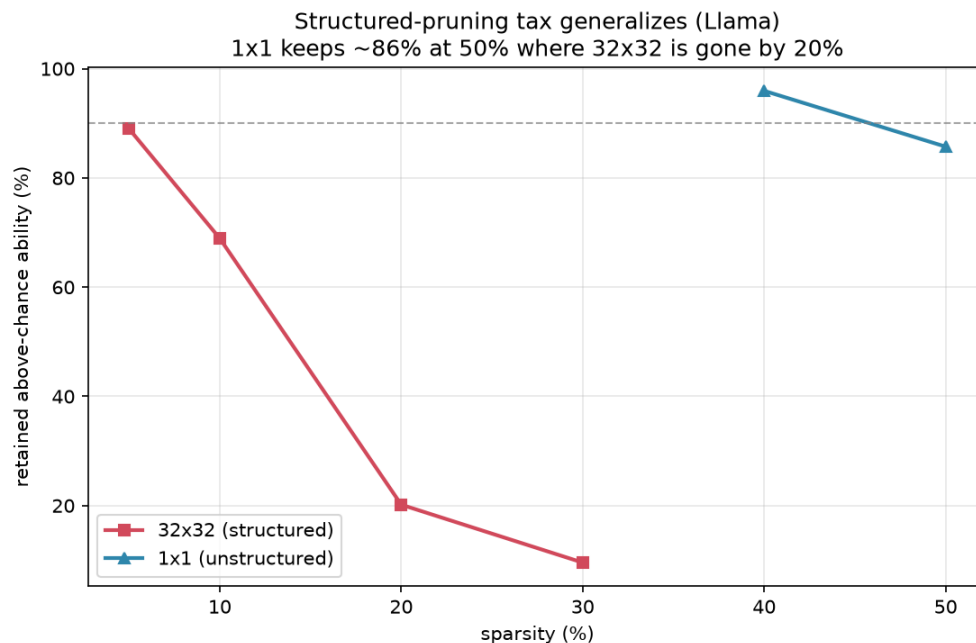
Result - structured ladder (F1), retained ability

model	p1	p2	p5	p10	p20	p30
Llama-3.2-3B-Inst	98	97	89	69	20	10
Qwen3-4B	100	96	90	79	43	23



Transferable - core findings generalize across both architectures

- The ~5 percent structured ceiling: Llama retains 89 percent at 5 percent vs Qwen 90 percent.
- Repair is the dominant lever: repair variants sit near dense on both models.
- The structured-pruning tax: Llama 1x1 keeps 86 percent at 50 percent while 32x32 is gone by 20 percent (1x1 ppl 10.6/11.0/12.0/14.7 at p20/30/40/50 vs 32x32 ppl 105 at p20) - even stronger than on Qwen.
- Calibrated selection beats random selection on downstream given repair (~5 pp on both): Llama wanda_recon 91 > sparsegpt_recon 89 > random_recon 86.



Not transferable - Qwen3-4B-specific (scope these to Qwen)

- Magnitude pruning does NOT catastrophize Llama: 21.9 ppl at 5 percent (functional) vs Qwen 3449 (destroyed).
- Calibrated selection does NOT lose to random on Llama - it beats random even without repair (no-repair ppl at 5 percent: sparsegpt 14.8 < wanda 16.6 < random 17.5 < magnitude 21.9), the opposite of Qwen.
- The perplexity-inversion (Finding A) does NOT occur on Llama - random_recon is worst on BOTH perplexity and capability, so the two axes agree.
- Llama collapses more steeply above the ceiling: 20 percent retained at p20 vs Qwen's 43 percent.

Reliability

Downstream-anchored on Llama; first-pass (3 seeds for F4, minimum doses). The DIVERGENT claims above deserve 2-3 more seeds before publication. Base-vs-instruct is matched (both instruct), so the divergences are an architecture-or-training-recipe effect, not a base/instruct artifact.

Reading these together

The robust, load-bearing claims - a ~5 percent structured ceiling, repair as the dominant lever, and a large but diffuse redundancy that only 1x1 can reach (the structured tax) - now hold across two architectures (n=2). Several of the more surprising Qwen behaviours - magnitude catastrophe, calibrated selection losing to a coin flip, and perplexity moving opposite to capability - are model-specific and do not generalize. Scoping them to Qwen makes the core contribution more credible, not less.

Open / not done

- The FINDINGS_clean write-up itself (author's prose) - unchanged by this addendum.
- 2-3 more seeds on Llama's divergent claims (magnitude, selection) before papering them.
- Reconcile the Finding C anisotropy metric.
- Optional generative / out-of-distribution validation (report next-step #5) - not run.